

Course Overview

[Topics and Objectives](#) learn stuff :)

V01 Introduction

Agenda

- Build Systems
 - With/Without Build System
- The Yocto Project
 - With/Without Build System
 - Architecture
 - Layers
 - Recipes
 - Adding an Application
 - Configuring a System
 - Building a bootable Image

What is embedded Linux?

Embedded Linux is the usage of the Linux Kernel and various Open-Source Components in embedded systems.

Advantages for Linux in Embedded

- Key advantage of Linux and open-source in embedded systems: Ability to re-use components
 - Many components for network protocols, multimedia, graphic, cryptographic libraries, etc. exist
 - Allows to quickly design and develop complicated products, based on existing components.
 - No-one should re-develop yet another operating system kernel, TCP/IP stack, USB stack or another graphical toolkit library.
- → Focus on the added value of your product.
- Low Cost! Free software can be duplicated on as many devices as you want, free of charge.
 - → No Software License! (except for GPL) ⇒ more details in lecture 09
 - Even the development tools are free, unless you choose a commercial embedded Linux edition.
- Allows to have a higher budget for the hardware or to increase the company's skills and knowledge

Benefits of a build system

With a build system

- Build systems automate the process of building a target system, including the kernel.
- Automatic download, configure, compile and install all the components in the right order, sometimes after applying.
- They make sure all the application dependencies are matched.
- The build becomes reproducible, which allows to easily change the configuration of some components, upgrade them, fix bugs, etc.

Without a build system

- Each application has to be built manually
- The root file system has to be created from scratch.
- The applications configurations have to be done by hand.
- Each dependency has to be matched manually.
- Integrating software from different teams is painful

joy

Comparison of distribution projects

- Buildroot
 - Simple to use.
 - Adapted for small embedded devices.
 - Not perfect if you need advanced functionalities and multiple machines support.
 - <http://buildroot.org/>
- OpenWRT
 - Based on Buildroot.
 - Primarily used for embedded network devices like routers.
 - <http://openwrt.org/>
- Poky
 - Part of the Yocto Project.
 - Using OpenEmbedded.
 - Suitable for more complex embedded systems.
 - Allows lots of customization.
 - Can be used for multiple targets at the same time.
 - <http://yoctoproject.org/>

Yocto

What is Yocto?

- The Yocto Project is a set of templates, tools and methods that allow to build custom embedded Linux-based systems.
- It is an open source project initiated by the Linux Foundation in 2010 and is managed by one of its fellows: Richard Purdie.